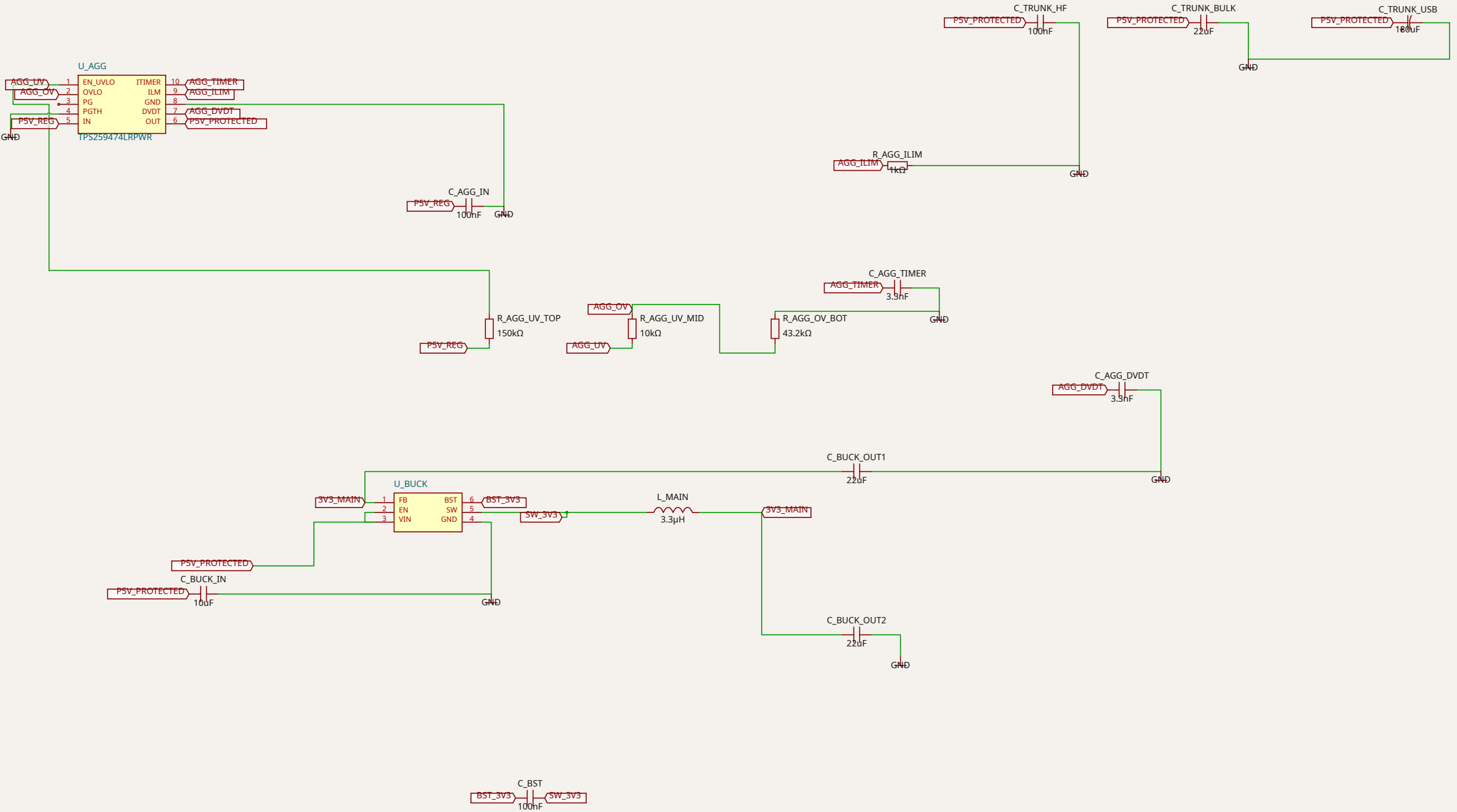




# USB\_CONTROLLED\_DEBUG\_HUB

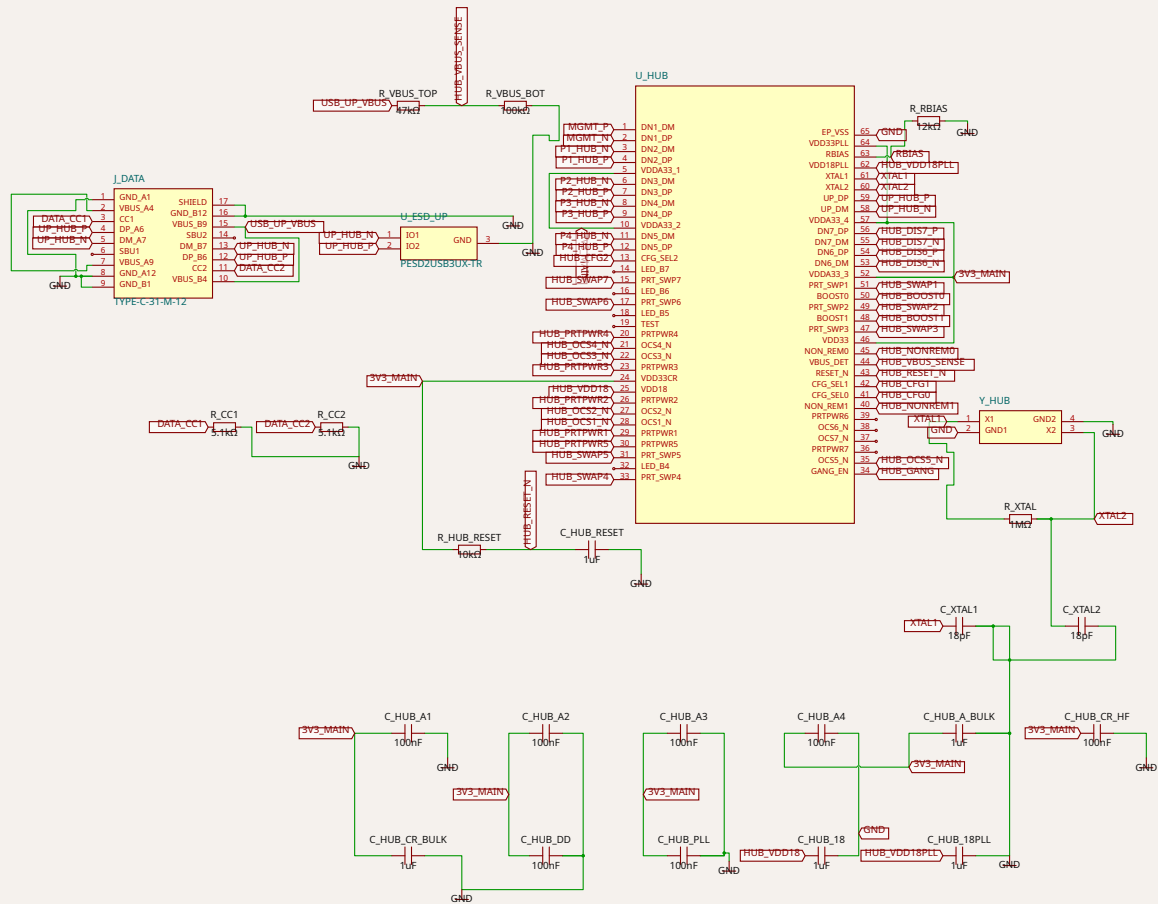
## POWER DISTRIBUTION — aggregate eFuse, USB bulk bank and 3.3 V buck



# USB\_CONTROLLED\_DEBUG\_HUB

## USB HUB — USB-C data-only upstream, ESD, USB2517I, straps, clock and bypass

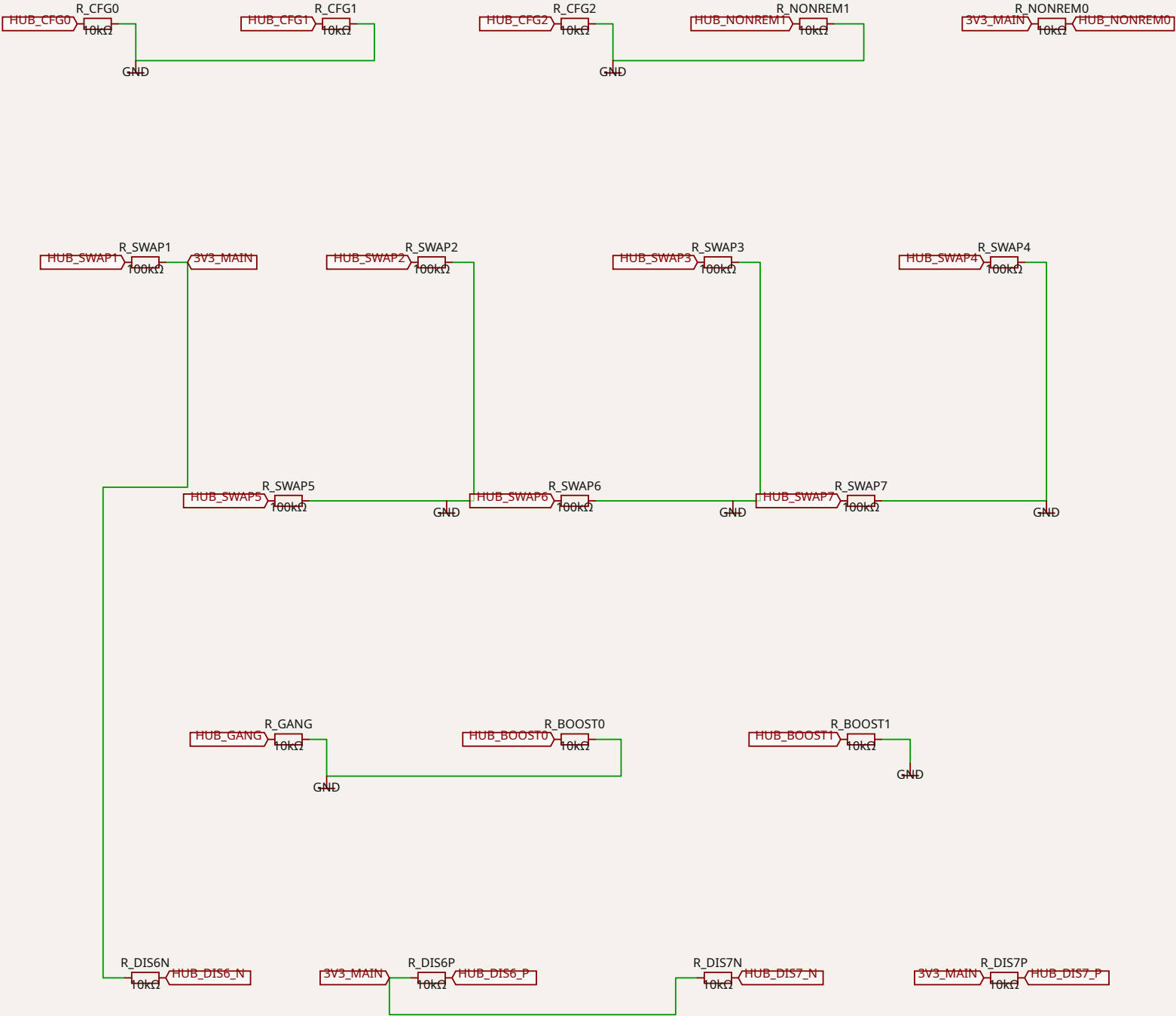
Page 3 of 10 • exact circuit.json SHA-256 6a187938ff9003c9... • 25 components • landscape page fit



INTENTIONAL NC: U\_HUB LED/TEST and disabled-port 6/7 PWR/OCS pins.

# USB\_CONTROLLED\_DEBUG\_HUB

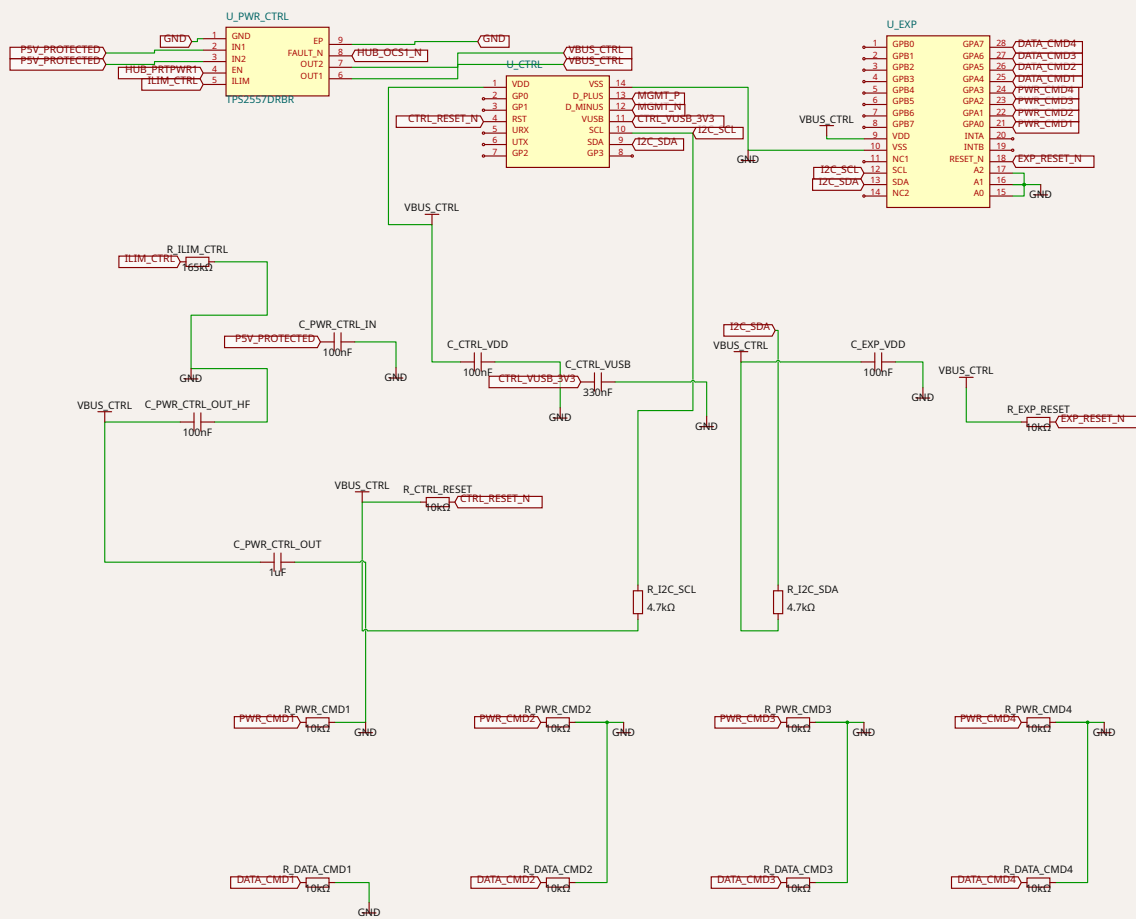
## USB HUB CONFIGURATION — P1 swapped, P2-5 normal, P6-7 disabled



# USB\_CONTROLLED\_DEBUG\_HUB

## MANAGEMENT — factory MCP2221A HID/I2C and MCP23017 command bank

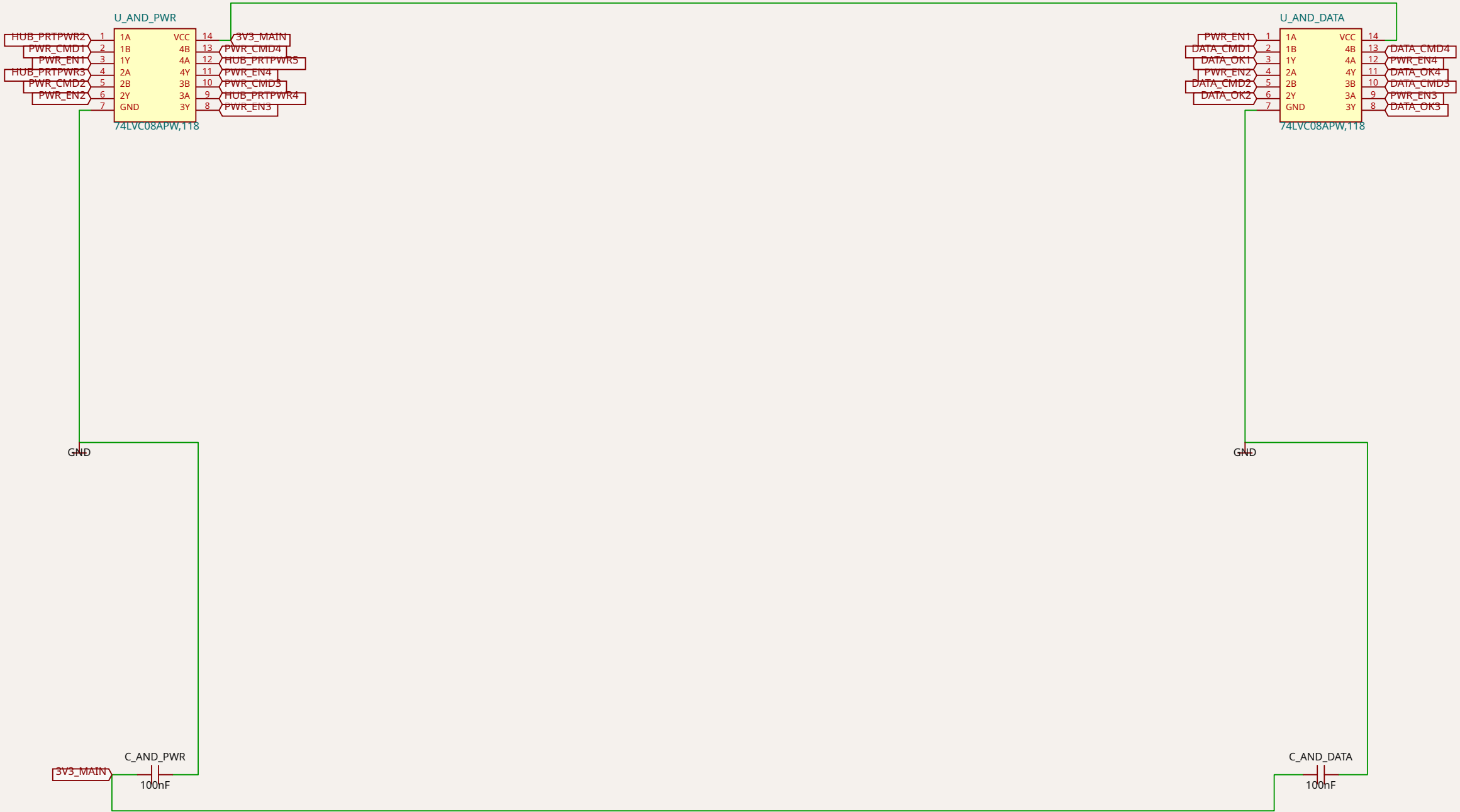
Page 5 of 10 • exact circuit.json SHA-256 6a187938ff9003c9... • 22 components • landscape page fit



INTENTIONAL NC: unused U\_CTRL GPIO/UART; U\_EXP GPB/INT/NC pins.

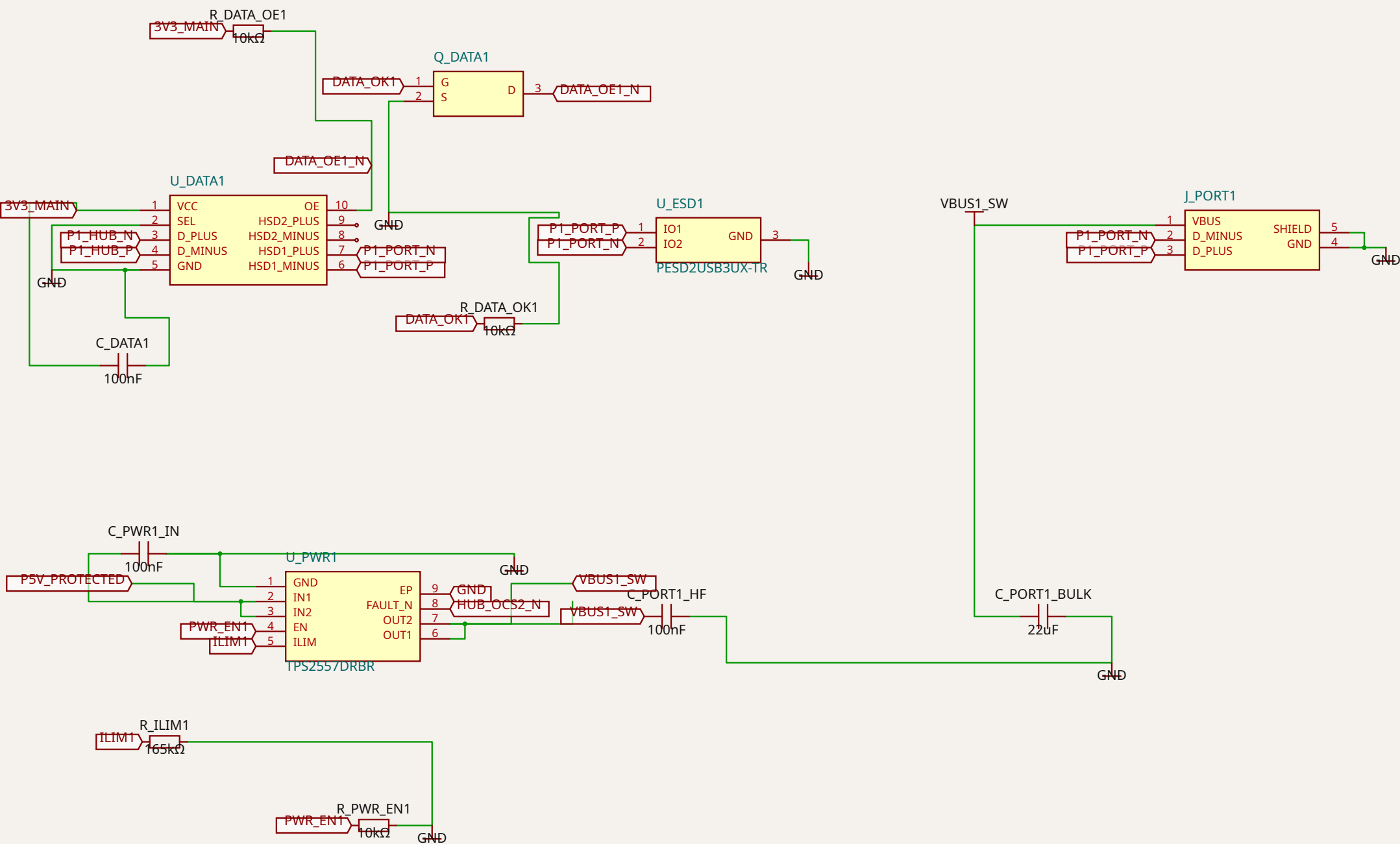
# USB\_CONTROLLED\_DEBUG\_HUB

## INTERLOCKS — hub policy AND host command; data follows commanded power enable



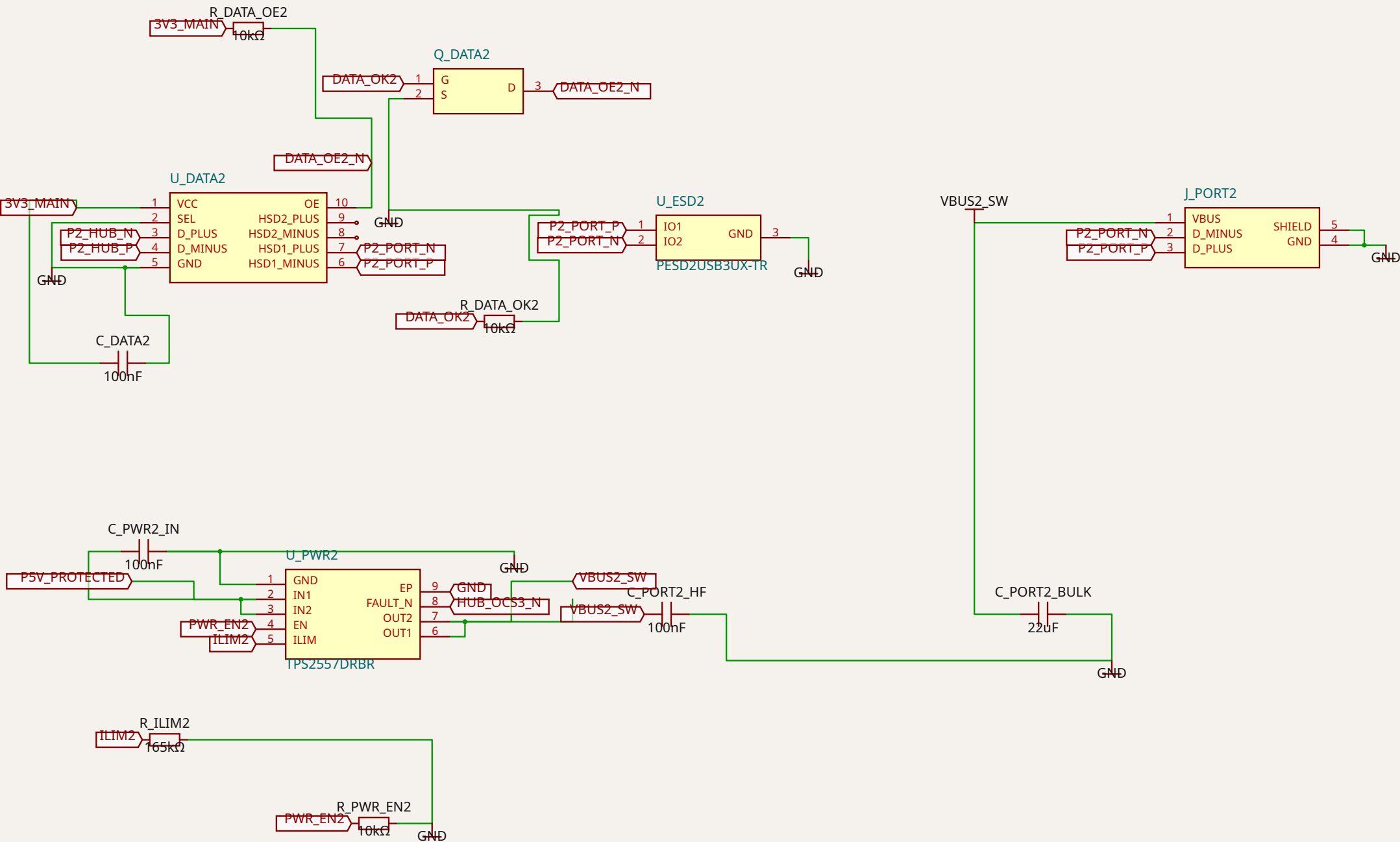
# USB\_CONTROLLED\_DEBUG\_HUB

## EXTERNAL PORT 1 — independent power/data disconnect



# USB\_CONTROLLED\_DEBUG\_HUB

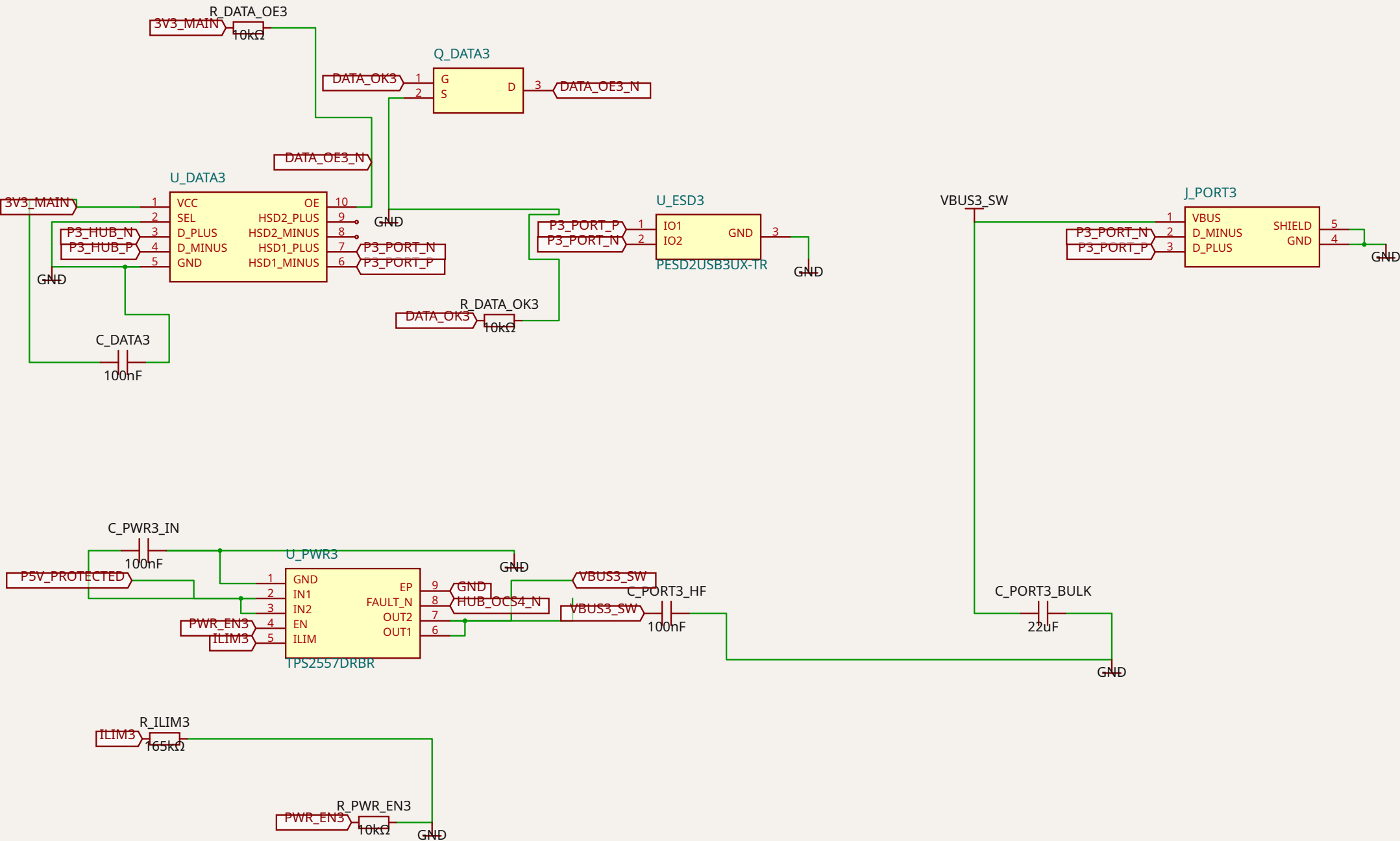
## EXTERNAL PORT 2 — independent power/data disconnect





# USB\_CONTROLLED\_DEBUG\_HUB

## EXTERNAL PORT 3 — independent power/data disconnect



# USB\_CONTROLLED\_DEBUG\_HUB

## EXTERNAL PORT 4 — independent power/data disconnect

